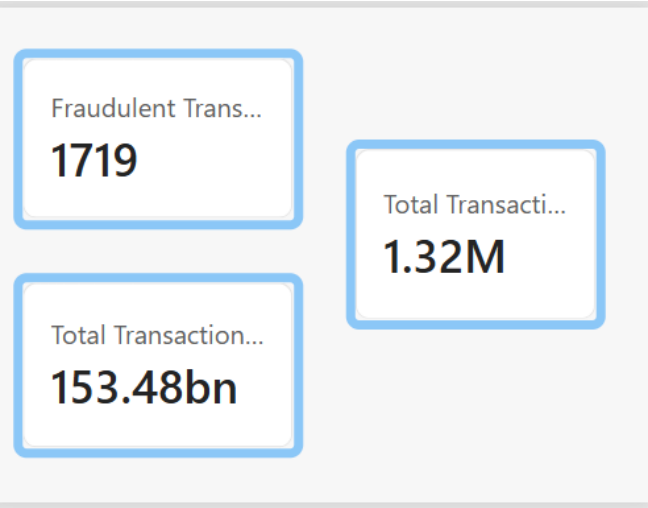
AI/ML Transaction & Fraud Analysis

Business Insight Report for the Transaction Monitoring Dashboard

Prepared by Vetri Selvan	**Report purpose** Summarise the visible transaction and fraud indicators, explain chart patterns and outline practical validation steps.

Scope

Three supplied dashboard screenshots. The report uses only visible labels and values; the underlying transaction-level dataset, date definitions and model evaluation measures were not supplied.



Executive Summary

The dashboard shows **1.32M total transactions**, **153.48bn total transaction amount**, **115.99K average transaction amount**, **2.80M total initial balance**, **10K accounts** and **1,719 fraudulent transactions**. These are the visible KPI card values; their currency, reporting period and aggregation definitions are not stated in the screenshots.

The fraud outcome donut labels the displayed transaction split as **False 99.87%** and **True 0.13%**, with a rounded center value of 1M. The separate KPI card reports 1.32M total transactions, so the rounded chart label should not be used to reconcile totals. Alert counts appear higher for cycle than fan_in. The timestamp line rises sharply at the start of its plotted sequence and then fluctuates around the mid-to-high 6,000s.

Headline KPIs - values visible in the screenshots

Average Transaction Amount	Total Initial Balance	Fraudulent Transactions
115.99K	2.80M	1719
Total Transaction Amount	**Total Accounts**	**Total Transactions**
153.48bn	10K	1.32M

The displayed fraud share is a screenshot label and does not establish model accuracy, recall, precision or the operational false-positive rate. Those measures require validated labels and transaction-level data.

1. Dashboard Overview

The dashboard combines six headline cards with four visuals: transaction volume by timestamp, alert count by alert type, transaction amount by fraud status and transaction count by fraud status. The second screenshot also shows filters for IS_FRAUD and ALERT_TYPE.

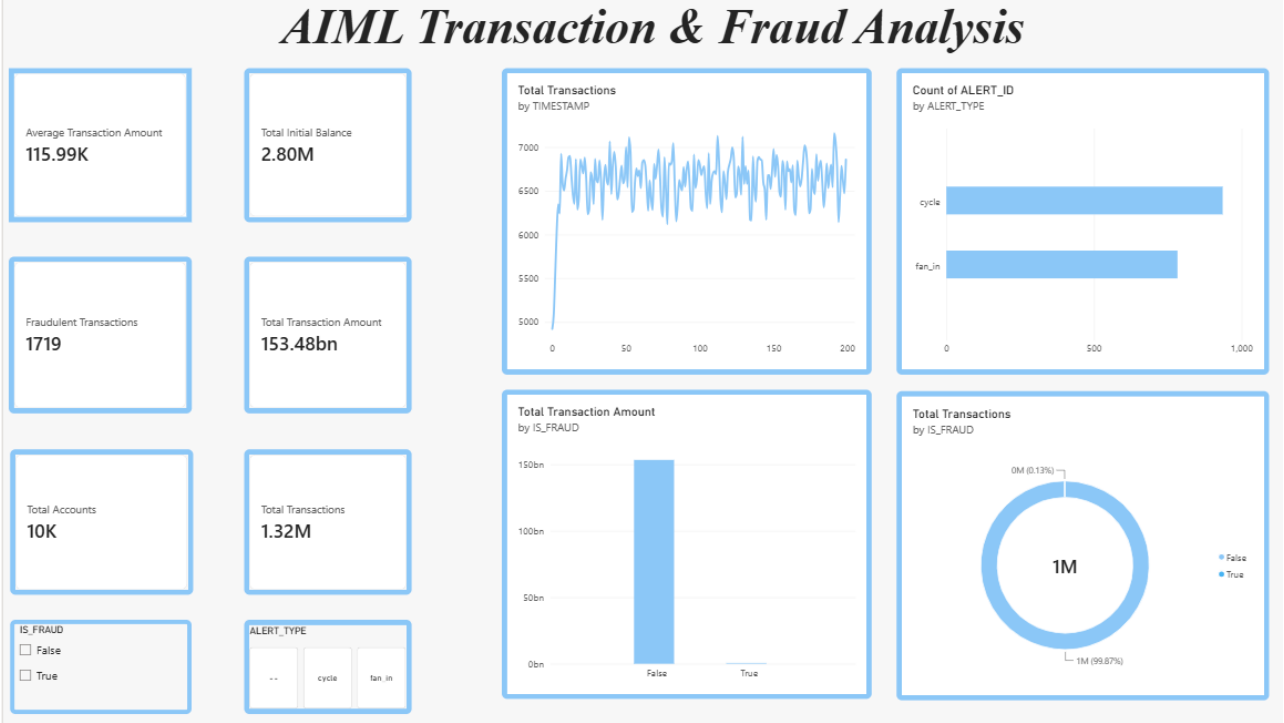


Figure 1. Full AI/ML transaction and fraud dashboard with KPI cards, charts and visible filters.

Read the dashboard from total volume and value to fraud outcomes and alert categories. The cards provide scale, while the charts show how the displayed values are distributed. Confirm filter selections before comparing figures across views.

How to use the dashboard

- Review transaction count and amount together to understand the scale represented.
- Compare the fraud count card with the fraud-status charts, and confirm both use the same filters.
- Use alert type to review the distribution of generated alerts.
- Check the timestamp field and reporting period before describing a trend.

2. KPI and Transaction Performance Analysis

The screenshots display six summary values. The metric labels and compact units are transcribed as shown; without the underlying data model, their precise calculation and period cannot be independently confirmed.

KPI	Displayed value	Interpretation / validation note
Average Transaction Amount	115.99K	Confirm currency, unit and whether the average includes all transaction statuses.
Total Initial Balance	2.80M	The screenshot does not define balance timing or the accounts included.
Fraudulent Transactions	1,719	Compare against the labeled outcome field and confirm the fraud definition.
Total Transaction Amount	153.48bn	Large aggregate amount; currency and period are not visible.
Total Accounts	10K	Confirm whether this is distinct accounts or a row count.
Total Transactions	1.32M	The fraud donut center shows a rounded 1M, so use the KPI label for the displayed total.

Reading the summary measures

The dashboard separates total transaction amount from average transaction amount and initial balance. These measures may use different units and aggregation rules. Do not derive average loss, fraud rate by value or account-level exposure from the screenshot alone. The 1,719 fraud count and 0.13% donut share may also represent different filter context or rounding; validate their definitions against the source model.

3. Transaction Activity and Alert Profile

Transaction volume by timestamp

The line chart starts near 4,900 on the first visible point, climbs rapidly, then fluctuates mostly between roughly 6,100 and 7,100 across the displayed sequence. The horizontal axis is shown as numeric positions from 0 to 200, not readable calendar dates. The shape supports a visual description of variation, but not a date-based trend or seasonality claim.

Alert type counts

The ALERT_ID bar chart contains two visible types, cycle and fan_in. Cycle has the longer bar, while fan_in is somewhat lower. The axis reaches 1,000, but exact labels are not printed, so the comparison should remain directional.



Figure 2. Transaction timeline, alert type counts and transaction amount by fraud status.

For operational use, verify whether ALERT_ID is unique per alert and whether the displayed counts cover a consistent time range.

4. Fraud Outcome and Transaction Amount

Transaction count by fraud status

The donut chart labels False at 99.87% and True at 0.13%. The center reads 1M after rounding. The separate Total Transactions card reads 1.32M; preserve both labels as shown and check the chart aggregation and filter context before reconciling them.

Transaction amount by fraud status

The amount chart is dominated by False, with a much smaller True bar on a scale that reaches about 150bn. The chart supports a large difference in displayed total amount by status, but the exact True amount is not labeled. It does not reveal per-transaction severity, prevented loss or model effectiveness.

What the screenshots do not show

No precision, recall, false-positive rate, threshold, confusion matrix, date range or labeled sample definition is visible. Use validated model metrics and a clearly documented evaluation window before making decisions about alert thresholds or model performance.

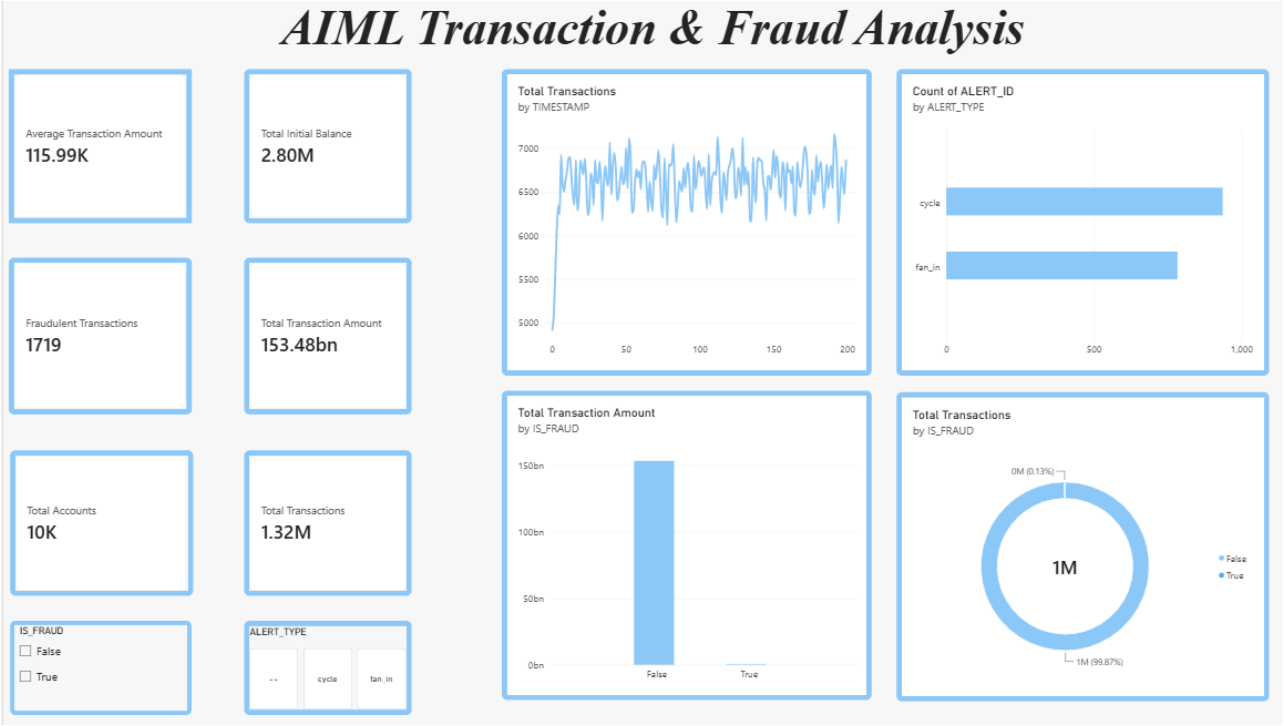


Figure 3. Dashboard view with the fraud outcome charts and IS_FRAUD / ALERT_TYPE filters.

5. Filters and Business Insights

The second screenshot displays filters for IS_FRAUD and ALERT_TYPE. Visible options include False and True, with cycle and fan_in shown as alert types. The active selections are not identifiable; this KPI set is treated as one displayed view.

Key observations

Observation	Business implication	Recommended action
The fraud status donut shows 99.87% False and 0.13% True.	The displayed classes are highly imbalanced.	Review precision, recall and precision-recall curves alongside the alert threshold.
The dashboard reports 1,719 fraudulent transactions.	This is a visible count, but its reporting period and labeling rule are unknown.	Confirm fraud labels, time window and whether the card counts distinct transactions.
Transaction amount by status is dominated by False.	Transaction count and amount may tell different risk stories.	Compare per-transaction amounts and total exposure by class using row-level data.
Cycle alerts have a longer bar than fan_in.	Alert mix may affect investigation workload.	Measure confirmed-fraud yield and review time by alert type.
The timestamp chart uses numeric sequence positions.	The plot does not establish calendar timing or seasonality.	Validate timestamp parsing, sorting, aggregation interval and missing periods.

Additional recommendations

- Document currency, time window, row grain and definitions for each KPI.
- Show exact counts for fraud outcomes and alert types, with the applied filters.
- Add model quality measures such as precision, recall, false-positive rate and PR-AUC.
- Monitor fraud rate and alert yield by time, account and transaction amount.
- Review category spelling and filter defaults so report exports are comparable.

Conclusion

The dashboard offers a compact view of transaction activity and fraud indicators. It reports 1.32M total transactions, 153.48bn total transaction amount and 1,719 fraudulent transactions. The status donut shows a 0.13% True share, while the amount chart is visually dominated by False. These values are transcribed from the dashboard and should be checked against the underlying model before operational use.

The alert chart shows more cycle than fan_in alerts, and the timestamp series varies across the visible sequence. The next review should confirm the date range, metric definitions, fraud label rules and model performance measures, then compare alert yield and value exposure under consistent filters.

Appendix: Data Notes

- This report is based on the three user-supplied screenshots; no transaction-level data file was included.
- Visible cards: Average Transaction Amount 115.99K; Total Initial Balance 2.80M; Fraudulent Transactions 1719.
- Visible cards: Total Transaction Amount 153.48bn; Total Accounts 10K; Total Transactions 1.32M.
- The fraud donut displays False 99.87%, True 0.13% and a rounded center label of 1M.
- Chart bar values and timestamp values are approximate visual readings where exact labels are not printed.
- The screenshots do not identify currency, reporting window, KPI formulas or model evaluation metrics.

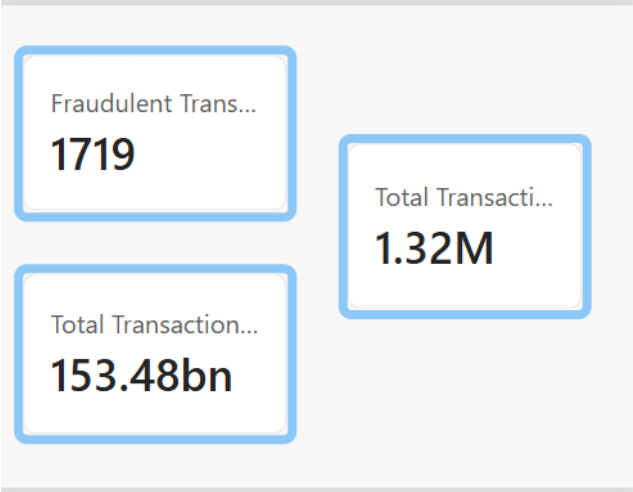


Figure 4. KPI cards for transaction amount, initial balance, fraud count and transaction total.